# Nada Ahmed

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## Summary

Computer Science graduate with Honours specializing in AI/ML, intelligent agents, and LLM systems. Built 15+ projects, including award-winning blockchain payroll and facial-recognition systems, with internship experience at SAIB Bank and Dell Technologies.

## Education

**Arab Academy for Science, Technology & Maritime Transport** **Sep 2022 – Jun 2026**
*Bachelor of Computer Science (Dual Degree), Graduated with Honours* – Cumulative GPA: 3.69 / 4.00 *Cairo, Egypt*

- Dual-degree program in collaboration with the University of Northampton (UK).
- Coursework: Machine Learning, Computer Vision, NLP, Algorithms & Optimization, Parallel Computing, Compilers, Databases, Robotics Applications, Software Engineering (SDLC).

## Skills

**Programming Languages:** Python, Java, C/C++, Go, JavaScript, TypeScript, Dart, PHP, R, Bash, SQL, HTML/CSS
**AI / ML & Data:** TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, OpenCV, Librosa, HuggingFace Transformers, Whisper, Docling; PDF extraction, OCR, NLP, Generative AI & LLM fine-tuning (DeepSeek-Coder, FLAN-UL2), Computer Vision, Reinforcement Learning, Explainable AI (SHAP, LIME, Grad-CAM)
**Web & Application:** Flask, FastAPI, React, Tailwind CSS, shadcn/ui, Vite, JavaFX, Tkinter, Flutter, SwiftUI, PyWebView
**Blockchain & Security:** Hyperledger Fabric, Chaincode (Go), Privy MPC / threshold custody, X.509 PKI, Cryptography
**Data, Cloud & DevOps:** MySQL, AWS RDS, SQLite, Docker, Kubernetes, Git, GitHub Actions (CI/CD), launchd
**CS Foundations:** Algorithm design & optimization, Data structures, Parallel computing, Compilers (lexing/parsing), Robotics & embedded (Arduino), SDLC
**APIs & Tools:** Twilio, Groq, Pydantic, Pytest, Graphviz, LaTeX
**Soft Skills:** Team collaboration, Analytical thinking, Communication, Mentorship

## Experience

**SAIB Bank** **Aug 2026 – Sept 2026**
*FinTech Intern* *Cairo, Egypt*

- Studied financial technology under the Head of FinTech & Startup Partnerships and completed an emerging-technology case study.

**Dell Technologies** **Aug 2025 – Sept 2025**
*Summer Intern* *Cairo, Egypt*

- Gained cross-department insight through sessions with Dell employees across sales, pre-sales, and engineering roles.
- Completed workshops on front-end development, containerization (Docker, Kubernetes), and software engineering practices.
- Applied design thinking to enhance Dell's onboarding experience; shadowed a project manager and software engineer to learn daily workflows.
- Led the Academy Project from concept to business proposal, strengthening technical and collaboration skills.

**DEPI – Microsoft ML Engineering Curriculum** **Nov 2024 – May 2025**
*AI/ML Engineer Intern* *Zamalek, Egypt*

- Completed a 7-month intensive AI/ML engineering program (Microsoft ML Engineering curriculum) covering NLP, generative AI, computer vision, and model deployment.
- Executed hands-on projects at each milestone and an end-to-end capstone (Jarvis AI Assistant) integrating multimodal workflows and intelligent agent orchestration.

**Megz** **Aug 2020 – Nov 2021**
*Computer Science Instructor* *Cairo, Egypt*

- Assisted 30+ students with programming fundamentals, improving proficiency in Python and pseudocode.
- Supported the lead teacher in managing classrooms, grading, and tracking student progress; delivered personalized guidance that improved small-group performance by 20%.
- Mentored students in debugging and problem-solving, fostering a collaborative learning environment.

## Projects

***Graduation Project***

**Payramid: Money Transaction Microservice** **2025 – 2026**

*Go, Hyperledger Fabric, Privy MPC, Twilio, TypeScript*

- Built a blockchain-based payroll-orchestration and immutable-audit layer enabling employers to disburse salaries instantly with cryptographically verifiable proofs of payment, settling on Egypt's national rails (InstaPay, Meeza, ACH) within banking (Law 194/2020) and data-protection (Law 151/2020) regulation.
- Engineered the smart-contract layer as Hyperledger Fabric chaincode (Go) on a 3-organization RAFT network with `MAJORITY` endorsement; validated with an automated Go suite of 91 passing tests (100%) covering serialization integrity, double-spend prevention, and a full STRIDE threat model.
- Delivered offline reach via encrypted SMS relay (one-time withdrawal codes for feature phones) and non-custodial wallets using Privy threshold (MPC) key management; authored the complete IEEE 830 SRS and IEEE 829 test plan.

***Artificial Intelligence & Machine Learning***

**Financial Reports Analysis** | *Python, FastAPI, Docling, Docker* **Sept – Oct 2026**

- Built a local application that locates and extracts statement pages from annual and interim financial-report PDFs, preserving source-page provenance and structuring periods, currencies, scales, and labels.
- Implemented deterministic `Decimal` metrics and accounting identity checks; linked figures to highlighted source regions and held inconsistent statements for review with explicit reasons.
- Delivered a background-processing workflow, fixed-template summaries limited to eligible facts, issuer-separated evaluation gates, and native macOS and Docker run profiles.

**Jarvis AI Assistant** | *Python, DeepSeek-Coder, FLAN-UL2, HuggingFace Transformers, Whisper* **Feb 2025 – On hold**

- Built a modular Jarvis-style AI assistant for code generation, image captioning, and voice interaction, with prompt-based instruction tuning and custom toolchain integration for autonomous task execution.
- Fine-tuned DeepSeek-Coder and FLAN-UL2 for multimodal encoder-decoder workflows; integrated NLP, Whisper-based speech recognition, and intelligent agent orchestration.

**Oribu** | *Python, OpenCV, Siamese Neural Network, Twilio API* **Jan – Jul 2023**

- Built an AI facial-recognition system using a Siamese neural network with contrastive loss for high-accuracy one-shot face verification, designed for criminal detection.
- Implemented real-time recognition with automated SMS alerts via the Twilio API. *(First Place AI/ML Project, AASTMT.)*

**Music Genre Detection** | *Python, TensorFlow/Keras, Librosa, Flask* **Dec 2025**

- Designed a CNN-LSTM-Attention deep-learning model classifying audio into 10 genres from mel-spectrograms, served through a Flask web interface.
- Added explainable-AI tooling (SHAP, LIME, Grad-CAM, Integrated Gradients), unit tests, and Sphinx-based API documentation.

**CMAPSS Analytics Assistant** | *Python, FastAPI, Groq Llama 3.3 70B* **May 2026**

- Built a NASA CMAPSS turbofan-degradation analytics service that parses FD001–FD004 telemetry, computes Remaining Useful Life (RUL), and ranks sensors by stability and informativeness.
- Exposed a FastAPI backend with an LLM-powered natural-language QA endpoint over failure-prediction signals.

**Fera5 Invaders** | *Python, Pygame, NumPy, TensorFlow, PyTorch, Scikit-learn* **Nov 2024 – Feb 2025**

- Developed an arcade-style space-shooter with a reinforcement-learning agent that autonomously learns to play the game.

***Systems, Algorithms & Security***

**Parallel FFT Image Processing** | *C++, CMake, Parallel Computing* **Jan 2026**

- Implemented 2D FFT-based image processing in C++ with sequential, partially-parallel, and fully-parallel variants to study parallel speedup, benchmarked through an automated CMake harness.

**Interactive Knapsack Planner** | *C++, SFML 3.0, Dear ImGui* **May 2025**

- Built an interactive algorithm visualizer for the 0/1 knapsack problem (greedy, brute-force, dynamic programming, branch-and-bound) with live DP-table animation and performance metrics.
- Added a Huffman-encoding module visualizing tree construction, bit-level encoding/decoding, and character-frequency analysis.

**Simple Certificate Authority** | *Python, Cryptography, X.509 PKI* **May 2026**

- Built a modular X.509 Certificate Authority supporting root-CA init, issuance, revocation (CRL), verification, renewal, and PKCS#12 export, with passphrase-encrypted keys and atomic database writes.

**Text Editor (SCLPL)** | *Python, Tkinter, Pillow, Graphviz, Pytest* **Nov 2024 – Feb 2025**

- Built a multi-tab editor with a full lexical and syntax-analysis pipeline for a custom toy language (SCLPL), including a hand-written lexer and parser for educational tooling.
- Added automatic language recognition, syntax highlighting, and parse-tree visualization.

**OverClocked** | *Bash, Python, Zenity, Docker* **Dec 2024 – Feb 2025**

- Built a cross-platform (Ubuntu/macOS) system-monitoring tool for hardware/software performance metrics with logging, visualization, and containerized deployment.

***Applications, Automation & Games***

**Pixel Paradise** | *JavaScript, PHP, HTML/CSS, Python* **May 2024 – Feb 2025**

- Worked as a core developer on a team-built retro web arcade with playable browser games and a custom pixel-art interface.
- Contributed across gameplay and full-stack site features, including user accounts and score tracking.

**Personal Portfolio** | *React, TypeScript, Vite, CSS Modules* **2026**

- Built a responsive project and experience portfolio with light/dark themes, a filterable project archive, and a downloadable CV; deployed through GitHub Actions to GitHub Pages with a custom DNS hostname.

**Job Apply Bot** | *Python, Groq Llama 3.3 70B, tectonic, launchd* **2026**

- Built a compliant multi-ATS job-search automation that polls public APIs (Greenhouse, Lever, Ashby, Adzuna), filters new-grad SWE/ML roles by skill overlap and location, and logs matches to CSV.
- Auto-tailors a LaTeX resume per posting via an LLM and compiles per-job PDFs with tectonic; scheduled hourly via launchd.

**Autonomous Pick-and-Place Robot** | *Arduino (C++), Flutter, Bluetooth* **2026**

- Built the control system for an autonomous mobile robot (Arduino Uno) with differential drive, a 3-DOF servo arm/gripper, and ultrasonic + line-tracking sensors for a Pick-and-Place competition.
- Developed a Flutter Android app communicating over Bluetooth (HC-05) for manual and automatic navigation modes.

**Trivia Game** | *Java, JavaFX, AWS RDS* **Apr 2024 – Feb 2025**

- Built a trivia game with single-player and real-time multiplayer modes backed by an AWS-hosted database for room management, question delivery, and live score tracking.
- Implemented a multiplayer waiting room with real-time chat, "is typing" indicators, and audio/visual feedback cues using FXML interfaces.

**Jarvis Desktop (Fluid Jarvis Whisper)** | *React, TypeScript, Tailwind, Flask, PyWebView* **May 2025**

- Packaged the Jarvis assistant as an offline cross-platform desktop app with a React/Tailwind front end, Flask backend, and PyWebView wrapper built via Vite.

## Certifications, Awards & Training

**3rd Place, Arab FinTech Challenge 2026** – Jordan FinTech Festival, Jordan (Sept 2026)
Represented Egypt with support from the Central Bank of Egypt (CBE); competition organized by GIE.
**2nd Place, FinTech Got Talent 2026** – Egypt (Sept 2026)
Won with Payramid as part of a team sponsored by SAIB Bank; organized by FinTech Egypt under the CBE.
**First Place, AI/ML Project Submission** – AASTMT
**Microsoft ML Engineer Course** – DEPI (Digital Egypt Pioneers Initiative)
**AI & Machine Learning Course** – Universitat Autònoma de Barcelona
**Computer Science Camp** – Carnegie Mellon University

## Languages

**Arabic:** Native **English:** Fluent **Spanish:** Intermediate